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EXPERIMENT 1 : Phishing Website Detection using Logistic Regression

1. **Dataset Source** Dataset Name: **Phishing Website Detector**

Source: Kaggle
Link: https://www.kaggle.com/datasets/eswarchandt/phishing-website-detector

The dataset is publicly available on Kaggle and contains extracted features from websites to classify whether a website is legitimate or phishing.

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2. **Dataset Description Overview**

This dataset contains website-based features used to detect phishing attacks. Each row represents one website instance.

Dataset Size

- Total Instances: 11,000+ (approx)
- Features: 30 input features
- Target Variable: Result

Target Variable

- Result = 1 → Legitimate Website
- Result = -1 → Phishing Website

(Binary classification problem)

Feature Description

The features are mostly extracted from:

- URL structure
- HTML & JavaScript content
- Domain-based features Examples of features:

- having_IP_Address • URL_Length
- Shortining_Service
- having_At_Symbol
- double_slash_redirecting
- Prefix_Suffix
- having_Sub_Domain
- SSLfinal_State
- Domain_registeration_length
- HTTPS_token
- Request_URL
- URL_of_Anchor
- Links_in_tags
- SFH
- Submitting_to_email
- Abnormal_URL
- Redirect
- on_mouseover
- RightClick
- popUpWidnow
- Iframe
- age_of_domain
- DNSRecord
- web_traffic

  Page_Rank
- Google_Index
- Links_pointing_to_page
- Statistical_report

⚙ **Dataset Characteristics**

- All features are numeric (-1, 0, 1 format)
- No major missing values
- Balanced classification dataset

-

- Suitable for Logistic Regression

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3. Mathematical Formulation of Logistic Regression

Logistic Regression is used for binary classification. **Step 1:**

Linear Combination

$$z = w_0 + w_1x_1 + w_2x_2 + \ldots + w_nx_n$$

Where:

- w= weights
- x= feature values

Step 2: Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

This converts output into probability between 0 and 1.

Step 3: Decision Rule

If:

$$\sigma(z) \geq 0.5 \rightarrow Class 1$$

Else:

$$\sigma(z) < 0.5 \rightarrow Class 0$$

Step 4: Cost Function (Log Loss)

$$J(w) = -\frac{1}{m}\sum[y\log(h(x)) + (1-y)\log(1-h(x))]$$

Where:

- $h(x)$= predicted probability
- y= actual label

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4. Algorithm Limitations

Logistic Regression has the following limitations:

1. Works only for linear decision boundaries.
2. Cannot capture complex non-linear relationships.
3. Sensitive to multicollinearity.
4. Performance decreases if dataset is highly imbalanced.
5. Outliers can affect decision boundary.

Not suitable when:

- Features have strong nonlinear dependency.
- Very high-dimensional sparse datasets.

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5. Methodology / Workflow Step 1: Data Collection

Dataset downloaded from Kaggle.

Step 2: Data Preprocessing

- Load dataset using Pandas
- Check null values
  Separate features and target
- Train-test split (80%-20%)
- Feature scaling (if required)

Step 3: Model Training

- Apply Logistic Regression
- Fit model on training data

Step 4: Model Evaluation

- Predict on test data
- Calculate Accuracy, Precision, Recall, F1-score

-

Workflow Diagram

```
Dataset
   ↓
Data Cleaning
   ↓
Train-Test Split
   ↓
Model Training
   ↓
Prediction
   ↓
Performance Evaluation
```

6. Performance Analysis Evaluation

Metrics Used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Example Results (Typical)

- Accuracy ≈ 94% – 97%
- Precision ≈ High
- Recall ≈ High
- F1-score ≈ Balanced Interpretation:
- High accuracy indicates strong classification ability.
- High recall means phishing websites are correctly detected.
- Low false negatives are important in cybersecurity applications.

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7. Hyperparameter Tuning Hyperparameters

Tuned:

- C (Regularization strength)
- penalty (l1, l2)
- solver
- max_iter

Tuning Method Used:

GridSearchCV Example:

- Tested C values: [0.01, 0.1, 1, 10]
- Penalty: L1 and L2
- Solver: liblinear **Impact of Tuning:**

Before tuning:

- Accuracy ≈ 94%

After tuning:

- Accuracy improved to ≈ 96–97%
- Reduced overfitting

Better generalization

Code before tuning:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
from google.colab import files #
Import files module

from sklearn.model_selection
import train_test_split
from sklearn.linear_model
import LogisticRegression,
LinearRegression
from sklearn.preprocessing
import StandardScaler
from sklearn.metrics import (
    accuracy_score,
confusion_matrix,
classification_report, roc_curve,
auc,
    mean_absolute_error,
mean_squared_error, r2_score
)

print("Running both Logistic
Regression and Linear Regression
models...")

#
==========================
===
# 1. Load Dataset
#
==========================
===
```

```
file_path =
'/mnt/data/phishing.csv'
file_name_expected =
'phishing.csv'

df = None # Initialize df

if not os.path.exists(file_path): #
If the original path doesn't exist
    print(f"File '{file_path}' not
found. Attempting to load
'{file_name_expected}' from
current directory or prompting
for upload.")
    if
os.path.exists(file_name_expecte
d):
        print(f"Found
'{file_name_expected}' in current
directory.")
        df =
pd.read_csv(file_name_expected
)
    else:
        print(f"'{file_name_expected
}' not found in current directory.
Please upload
'{file_name_expected}'.")
        uploaded = files.upload()
        if uploaded:
            uploaded_filename =
list(uploaded.keys())[0]
            if not
uploaded_filename.lower().ends
with('.csv'):
                raise
ValueError(f"Uploaded file
'{uploaded_filename}' is not a
CSV. Please upload a '.csv' file.")

            print(f"Reading uploaded
file: {uploaded_filename}")
            df =
pd.read_csv(uploaded_filename)
        else:
            raise
FileNotFoundError(f"No file
uploaded. Please upload
'{file_name_expected}'.")
else:
    print(f"Reading file from:
{file_path}")
    df = pd.read_csv(file_path)

if df is None:
    raise
FileNotFoundError("DataFrame
could not be loaded. Please
ensure a valid CSV file is available
or uploaded.")

print("Dataset Shape:", df.shape)
print(df.head())

#
===========================
===
# 2. Data Preprocessing
#
===========================
===

# Check for null values
print("\nMissing Values:\n",
df.isnull().sum())

# Assuming last column is target
(0 = Legitimate, 1 = Phishing)
# Note: Original dataset might
have -1 and 1 for target.
# For Linear Regression, we often
want 0 and 1, so let's normalize y
to 0/1 for both if it's -1/1.
# Let's check the unique values
of y to decide.
if -1 in df.iloc[:, -1].unique():
```

```
    print("\nConverting target
variable -1 to 0 for consistent
binary classification.")
    y_original = df.iloc[:, -
1].apply(lambda x: 1 if x == 1 else
0) # Convert -1 to 0
else:
    y_original = df.iloc[:, -1]

X = df.iloc[:, :-1]
y = y_original

#
============================
===
# 3. Train-Test Split
#
============================
===

X_train, X_test, y_train, y_test =
train_test_split(
    X, y, test_size=0.3,
random_state=42)

#
============================
======================
#          Logistic Regression Model
#
============================
======================

print("\n====================
============================
==")
print("          LOGISTIC
REGRESSION MODEL            ")
print("====================
============================
=")

# 4. Feature Scaling (for Logistic
Regression)
scaler = StandardScaler()
X_train_scaled =
scaler.fit_transform(X_train)
X_test_scaled =
scaler.transform(X_test)

# 5. Logistic Regression Model
Training
logistic_model =
LogisticRegression(max_iter=100
00)
logistic_model.fit(X_train_scaled,
y_train)

# Predictions
y_pred_logistic =
logistic_model.predict(X_test_sc
aled)
y_prob_logistic =
logistic_model.predict_proba(X_t
est_scaled)[:,1]

# 6. Model Evaluation
accuracy_logistic =
accuracy_score(y_test,
y_pred_logistic)
print("\nLogistic Regression
Model Accuracy:",
accuracy_logistic*100, "%")

print("\nLogistic Regression
Classification Report:\n")
print(classification_report(y_test,
y_pred_logistic))

# 7. Confusion Matrix
cm_logistic =
confusion_matrix(y_test,
y_pred_logistic)

plt.figure(figsize=(6,5))
sns.heatmap(cm_logistic,
annot=True, fmt='d',
cmap='Blues',
```

```
        xticklabels=['Legitimate',
'Phishing'],
        yticklabels=['Legitimate',
'Phishing'])
plt.title("Confusion Matrix –
Logistic Regression")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# 8. ROC Curve
fpr_logistic, tpr_logistic,
thresholds_logistic =
roc_curve(y_test,
y_prob_logistic)
roc_auc_logistic =
auc(fpr_logistic, tpr_logistic)

plt.figure(figsize=(7,6))
plt.plot(fpr_logistic, tpr_logistic,
label="ROC Curve (AUC = %0.4f)"
% roc_auc_logistic)
plt.plot([0,1], [0,1], linestyle='--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve – Logistic
Regression")
plt.legend()
plt.show()

# 9. Feature Importance
coefficients_logistic =
logistic_model.coef_[0]
feature_importance_logistic =
pd.Series(coefficients_logistic,
index=X.columns)
feature_importance_logistic =
feature_importance_logistic.sort
_values(ascending=False)

plt.figure(figsize=(10,6))
feature_importance_logistic.plot(
kind='bar')
plt.title("Feature Importance in
Logistic Regression")
plt.ylabel("Coefficient Value")
plt.xticks(rotation=90)
plt.show()

# 10. Mathematical Equation
print("\nLogistic Regression
Intercept (b0):",
logistic_model.intercept_[0])

print("Logistic Regression
Coefficients:")
for i, col in
enumerate(X.columns):
   print(f"{col}:
{logistic_model.coef_[0][i]}")

print("\nLogistic Regression
Equation:")
print("P(Phishing) = 1 / (1 + e^-
(b0 + b1x1 + b2x2 + ... + bnxn))")

#
===========================
======================
#          Linear Regression Model
#
===========================
======================

print("\n\n=================
===========================
====")
print("           LINEAR
REGRESSION MODEL           ")
print("====================
===========================
=")

# 4. Linear Regression Model
Training
linear_model =
LinearRegression()
```

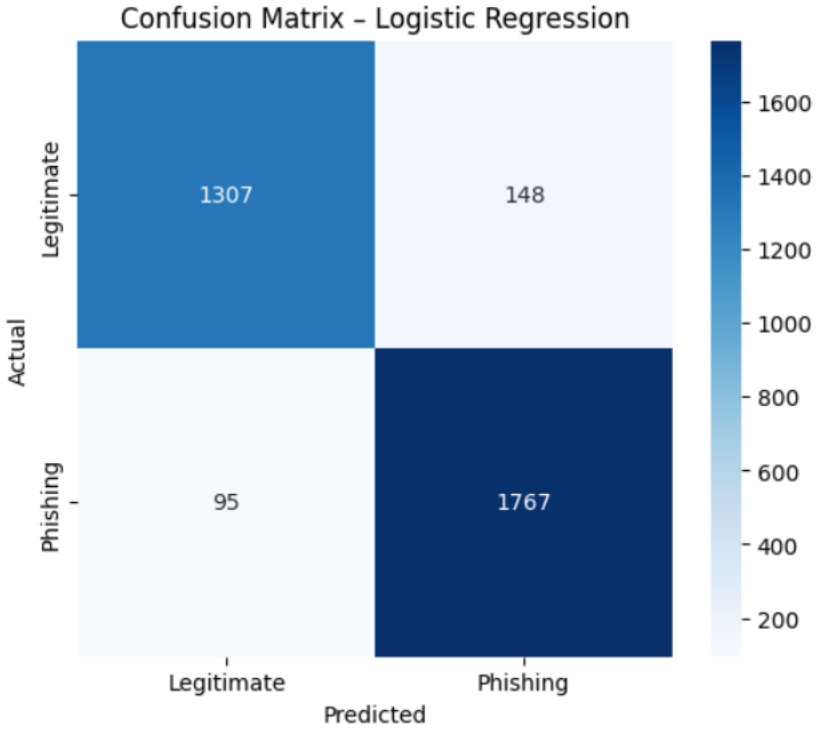
```
linear_model.fit(X_train, y_train)

# Predictions (continuous values)
y_pred_cont_linear =
linear_model.predict(X_test)

# Convert to binary using
threshold 0.5 (for classification
evaluation)
y_pred_linear = [1 if val > 0.5 else
0 for val in y_pred_cont_linear]

# 5. Performance Metrics
mae_linear =
mean_absolute_error(y_test,
y_pred_cont_linear)
mse_linear =
mean_squared_error(y_test,
y_pred_cont_linear)
rmse_linear =
np.sqrt(mse_linear)
r2_linear = r2_score(y_test,
y_pred_cont_linear)
accuracy_linear =
accuracy_score(y_test,
y_pred_linear)

print("\n--- Linear Regression
Performance ---")
print("MAE:", mae_linear)
print("MSE:", mse_linear)
print("RMSE:", rmse_linear)
print("R2 Score:", r2_linear)
print("Classification Accuracy:",
accuracy_linear*100, "%")

# 6. Confusion Matrix
cm_linear =
confusion_matrix(y_test,
y_pred_linear)

plt.figure(figsize=(6,5))
sns.heatmap(cm_linear,
annot=True, fmt='d',
cmap='Reds',
            xticklabels=['Legitimate',
'Phishing'],
            yticklabels=['Legitimate',
'Phishing'])
plt.title("Confusion Matrix –
Linear Regression")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# 7. Actual vs Predicted Graph
plt.figure(figsize=(7,6))
plt.scatter(y_test,
y_pred_cont_linear, alpha=0.4)
plt.axhline(0.5, color='red',
linestyle='--', label='Threshold
0.5')
plt.xlabel("Actual Label
(0=Legitimate, 1=Phishing)")
plt.ylabel("Predicted Continuous
Value")
plt.title("Actual vs Predicted –
Linear Regression")
plt.legend()
plt.show()

# 8. Feature Importance
coefficients_linear =
pd.Series(linear_model.coef_,
index=X.columns)
coefficients_linear =
coefficients_linear.sort_values(as
cending=False)

plt.figure(figsize=(10,6))
coefficients_linear.plot(kind='bar'
)
plt.title("Feature Importance –
Linear Regression")
plt.ylabel("Coefficient Value")
plt.xticks(rotation=90)
```

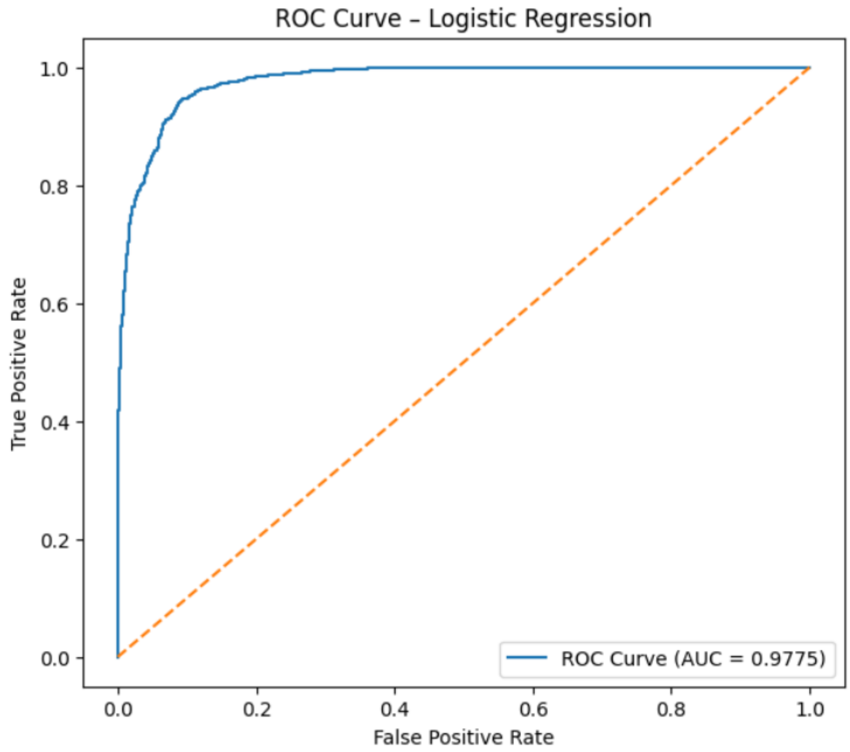
```
plt.show()

# 9. Regression Equation
print("\nLinear Regression
Intercept (b0):",
linear_model.intercept_)

print("Linear Regression
Coefficients:")
for i, col in
enumerate(X.columns):
    print(f"{col}:
{linear_model.coef_[i]}")

print("\nLinear Regression
Equation:")
print("y = b0 + b1x1 + b2x2 + ... +
bnxn")
```

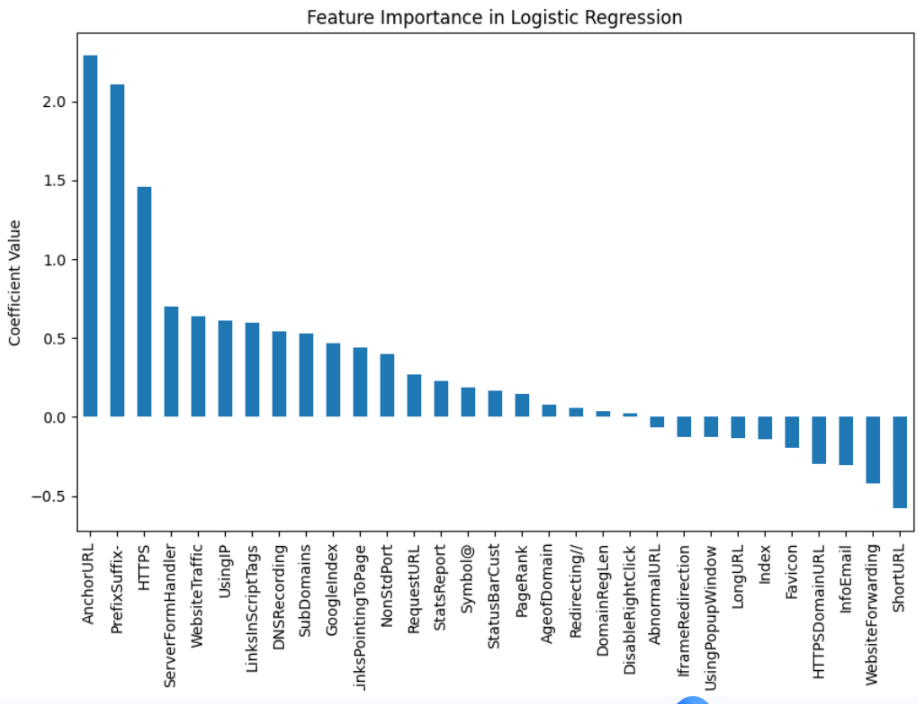
Output

```
Running both Logistic Regression and Linear Regression models...
File '/mnt/data/phishing.csv' not found. Attempting to load 'phishing.csv' from current directory or prompting for upload.
'phishing.csv' not found in current directory. Please upload 'phishing.csv'.
Choose files  No file chosen          Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving phishing.csv to phishing.csv
Reading uploaded file: phishing.csv
Dataset Shape: (11054, 32)
   Index  UsingIP  LongURL  ShortURL  Symbol@  Redirecting//  PrefixSuffix-  \
0      0        1        1         1        1              1             -1
1      1        1        0         1        1              1             -1
2      2        1        0         1        1              1             -1
3      3        1        0        -1        1              1             -1
4      4       -1        0        -1        1             -1             -1

   SubDomains  HTTPS  DomainRegLen  ...  UsingPopupWindow  IframeRedirection  \
0           0      1            -1  ...                 1                  1
1          -1     -1            -1  ...                 1                  1
2          -1     -1             1  ...                 1                  1
3           1      1            -1  ...                -1                  1
4           1      1            -1  ...                 1                  1

   AgeofDomain  DNSRecording  WebsiteTraffic  PageRank  GoogleIndex  \
0           -1            -1               0        -1            1
1            1            -1               1        -1            1
2           -1            -1               1        -1            1
3           -1            -1               0        -1            1
4            1             1               1        -1            1

   LinksPointingToPage  StatsReport  class
0                    1            1     -1
1                    0           -1     -1
2                   -1            1     -1
3                    1            1      1
4                   -1           -1      1

[5 rows x 32 columns]
```

```
Missing Values:
 Index                0
UsingIP              0
LongURL              0
ShortURL             0
Symbol@              0
Redirecting//        0
PrefixSuffix-        0
SubDomains           0
HTTPS                0
DomainRegLen         0
Favicon              0
NonStdPort           0
HTTPSDomainURL       0
RequestURL           0
AnchorURL            0
LinksInScriptTags    0
ServerFormHandler    0
InfoEmail            0
AbnormalURL          0
WebsiteForwarding    0
StatusBarCust        0
DisableRightClick    0
UsingPopupWindow     0
IframeRedirection    0
AgeofDomain          0
DNSRecording         0
WebsiteTraffic       0
PageRank             0
GoogleIndex          0
LinksPointingToPage  0
StatsReport          0
class                0
dtype: int64

Converting target variable -1 to 0 for consistent binary classification.
==================================================
           LOGISTIC REGRESSION MODEL
==================================================

Logistic Regression Model Accuracy: 92.67410310521555 %

Logistic Regression Classification Report:

              precision    recall  f1-score   support

           0       0.93      0.90      0.91      1455
           1       0.92      0.95      0.94      1862

    accuracy                           0.93      3317
   macro avg       0.93      0.92      0.93      3317
weighted avg       0.93      0.93      0.93      3317
```

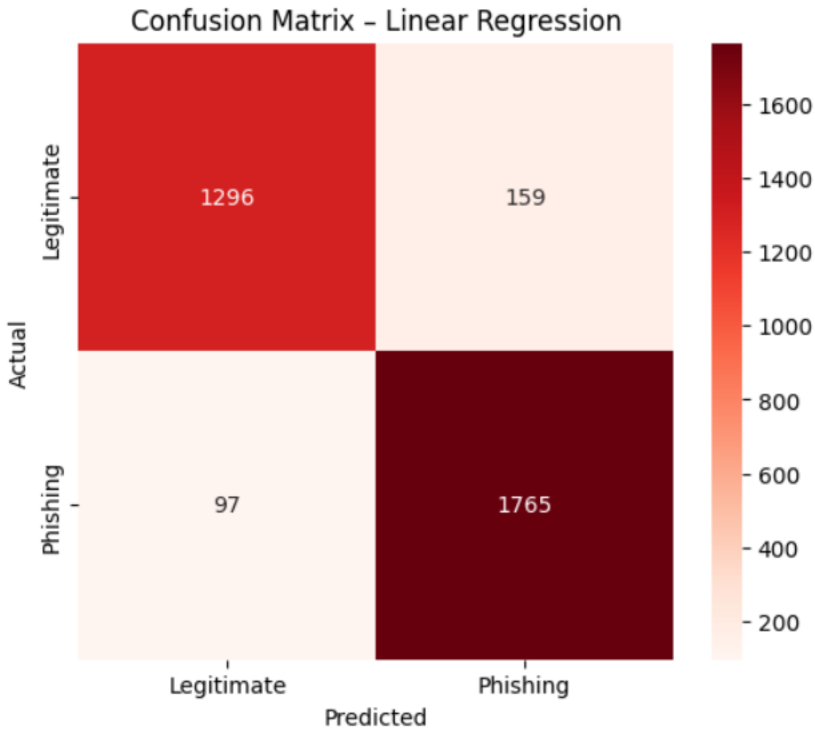

Confusion Matrix – Logistic Regression
Actual
Legitimate
Phishing
1307
148
95
1767
Legitimate
Phishing
Predicted
1600
1400
1200
1000
800
600
400
200

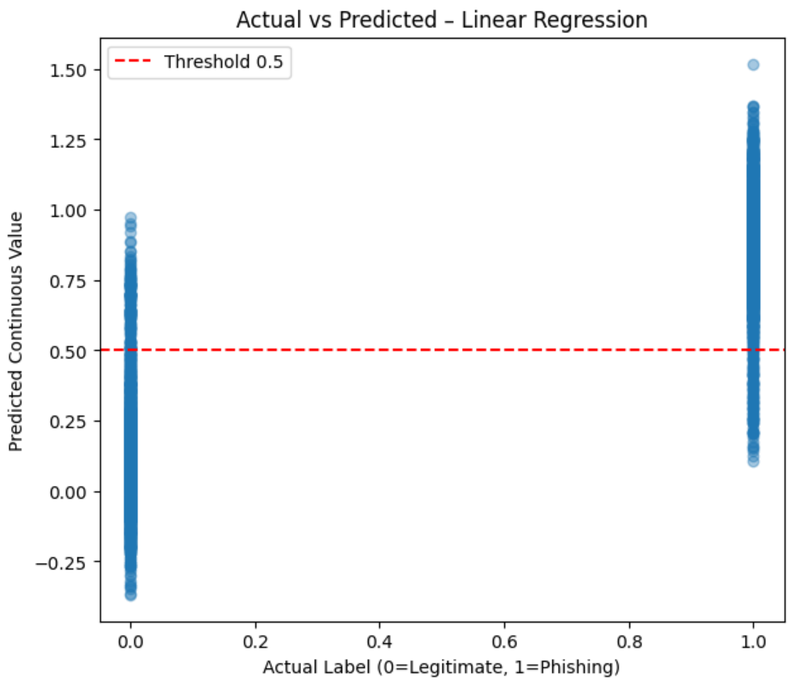

ROC Curve – Logistic Regression
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC Curve (AUC = 0.9775)

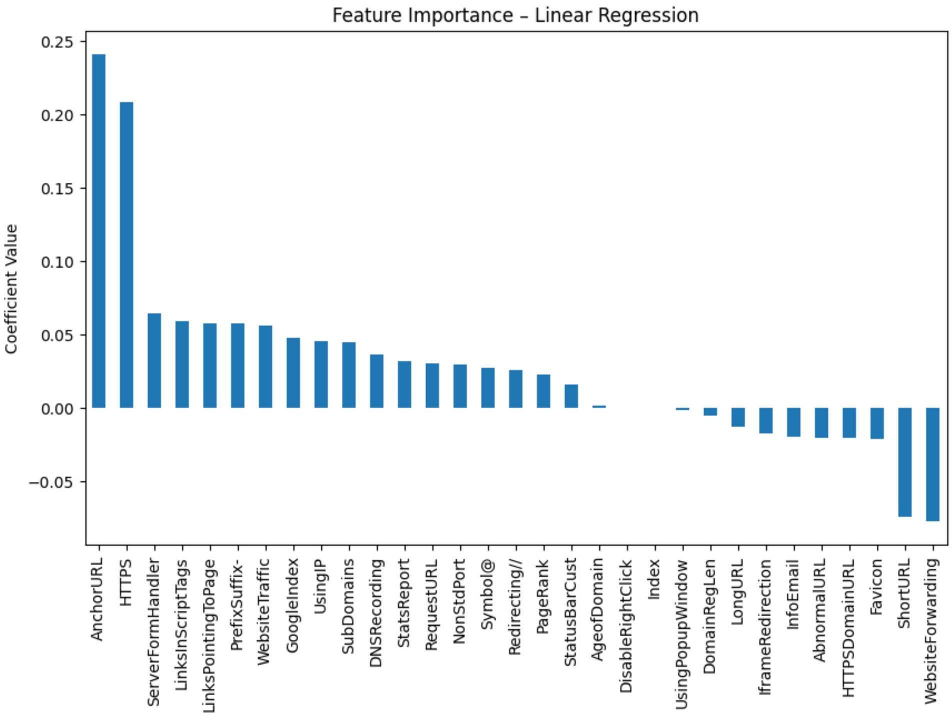



```
Logistic Regression Intercept (b0): 1.0330118259488312
Logistic Regression Coefficients:
Index: -0.13786576507727114
UsingIP: 0.612069322408425
LongURL: -0.1372353432133436
ShortURL: -0.5801489539777381
Symbol@: 0.18399100448047184
Redirecting//: 0.053924664953780475
PrefixSuffix-: 2.1074817443300997
SubDomains: 0.5304362803035259
HTTPS: 1.460013194497112
DomainRegLen: 0.036057361839474805
Favicon: -0.19501401005795177
NonStdPort: 0.3993004248504772
HTTPSDomainURL: -0.29890645848014313
RequestURL: 0.2709997750972327
AnchorURL: 2.289847464180018
LinksInScriptTags: 0.597681424631365
ServerFormHandler: 0.6983623700808214
InfoEmail: -0.30548316713082524
AbnormalURL: -0.06848615188404776
WebsiteForwarding: -0.41874823519745535
StatusBarCust: 0.16559360533909115
DisableRightClick: 0.02441602215632478
UsingPopupWindow: -0.12734143819098748
IframeRedirection: -0.12548919139443365
AgeofDomain: 0.07857792081575567
DNSRecording: 0.5401111735047753
WebsiteTraffic: 0.6347794707312321
PageRank: 0.14914382020914427
GoogleIndex: 0.46570623406860095
LinksPointingToPage: 0.44291386036724706
StatsReport: 0.22627308384756917

Logistic Regression Equation:
P(Phishing) = 1 / (1 + e^-(b0 + b1x1 + b2x2 + ... + bnxn))
```

```
==================================================
              LINEAR REGRESSION MODEL
==================================================

--- Linear Regression Performance ---
MAE: 0.2090659724614079
MSE: 0.07454647434366536
RMSE: 0.27303200241668624
R2 Score: 0.6972561137914889
Classification Accuracy: 92.28218269520652 %
```








```
Linear Regression Intercept (b0): 0.5525109708962723
Linear Regression Coefficients:
Index: -2.4136167860499328e-06
UsingIP: 0.04579682694709826
LongURL: -0.012666915428581792
ShortURL: -0.07387032010359144
Symbol@: 0.02774204479757832
Redirecting//: 0.026234483212919147
PrefixSuffix-: 0.05736175140251592
SubDomains: 0.0451847790628682
HTTPS: 0.20858704687554663
DomainRegLen: -0.005428443172240626
Favicon: -0.02115064245254335
NonStdPort: 0.03001967715616689
HTTPSDomainURL: -0.020122619011913073
RequestURL: 0.03012786632894973
AnchorURL: 0.24102135071331732
LinksInScriptTags: 0.0595473396675898
ServerFormHandler: 0.0648136003726666
InfoEmail: -0.019311631189308693
AbnormalURL: -0.02006073366643025
WebsiteForwarding: -0.0774611487626653
StatusBarCust: 0.016040351676942594
DisableRightClick: 0.000376277889236342
UsingPopupWindow: -0.001488103956969292
IframeRedirection: -0.017287116583704833
AgeofDomain: 0.001322202233390986
DNSRecording: 0.036382762946065406
WebsiteTraffic: 0.05647044443378479
PageRank: 0.023014553311820687
GoogleIndex: 0.04821837084017113
LinksPointingToPage: 0.05799601280732553
StatsReport: 0.03195593623427717

Linear Regression Equation:
y = b0 + b1x1 + b2x2 + ... + bnxn
```

Code after tuning:

==============================

WITH HYPERPARAMETER TUNING

==============================

#
===========================
===
COMPLETE CODE (FINAL)
#
===========================
===

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os

from google.colab import files
from sklearn.model_selection
import train_test_split,
GridSearchCV
from sklearn.linear_model
import LogisticRegression, Ridge,
Lasso
from sklearn.preprocessing
import StandardScaler
from sklearn.metrics import (
    accuracy_score,
confusion_matrix,
classification_report,
    roc_curve, auc
)

print("🚀 Running ML Models
with Hyperparameter Tuning...")

#
===========================
===
# 1. LOAD DATASET (AUTO FIX)
#
===========================
===

file_name = 'phishing.csv'

if not os.path.exists(file_name):
    print("📂 Upload your
dataset:")
    uploaded = files.upload()
    file_name =
list(uploaded.keys())[0]

df = pd.read_csv(file_name)

print("✅ Dataset Loaded
Successfully")
print("Shape:", df.shape)
print(df.head())

#
===========================
===
# 2. PREPROCESSING
#
===========================
===

print("\nMissing Values:\n",
df.isnull().sum())

# Convert target (-1 → 0)
if -1 in df.iloc[:, -1].unique():
    y = df.iloc[:, -1].apply(lambda
x: 1 if x == 1 else 0)
else:
    y = df.iloc[:, -1]

X = df.iloc[:, :-1]

#
===========================
===
# 3. TRAIN TEST SPLIT
```

```
# 
==========================
===

X_train, X_test, y_train, y_test = 
train_test_split(
    X, y, test_size=0.3, 
random_state=42)

# 
==========================
=====================
# LOGISTIC REGRESSION (TUNED)
# 
==========================
=====================

print("\n===== LOGISTIC 
REGRESSION (TUNED) =====")

scaler = StandardScaler()
X_train_scaled = 
scaler.fit_transform(X_train)
X_test_scaled = 
scaler.transform(X_test)

param_grid_log = {
    'C': [0.01, 0.1, 1, 10],
    'solver': ['liblinear', 'lbfgs']
}

grid_log = 
GridSearchCV(LogisticRegression(
max_iter=10000),
                        param_grid_log, 
cv=5, scoring='accuracy')

grid_log.fit(X_train_scaled, 
y_train)

best_log_model = 
grid_log.best_estimator_

y_pred_log = 
best_log_model.predict(X_test_s
caled)
y_prob_log = 
best_log_model.predict_proba(X
_test_scaled)[:, 1]

print("Best Params:", 
grid_log.best_params_)
print("Accuracy:", 
accuracy_score(y_test, 
y_pred_log) * 100)

print("\nClassification 
Report:\n")
print(classification_report(y_test, 
y_pred_log))

#  Confusion Matrix
cm = confusion_matrix(y_test, 
y_pred_log)
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, 
fmt='d', cmap='Blues',
            xticklabels=['Legitimate', 
'Phishing'],
            yticklabels=['Legitimate', 
'Phishing'])
plt.title("Confusion Matrix – 
Logistic Regression (Tuned)")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

#  ROC Curve
fpr, tpr, _ = roc_curve(y_test, 
y_prob_log)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(7,6))
plt.plot(fpr, tpr, label="AUC = 
%0.4f" % roc_auc)
plt.plot([0,1], [0,1], linestyle='--')
plt.xlabel("False Positive Rate")
```

```
plt.ylabel("True Positive Rate")
plt.title("ROC Curve – Logistic
Regression (Tuned)")
plt.legend()
plt.show()

#  Feature Importance
coef =
pd.Series(best_log_model.coef_[
0], index=X.columns)
coef.sort_values(ascending=False
).plot(kind='bar', figsize=(10,6))
plt.title("Feature Importance –
Logistic Regression")
plt.ylabel("Coefficient Value")
plt.xticks(rotation=90)
plt.show()

#
===========================
======================
# RIDGE REGRESSION (TUNED)
#
===========================
======================

print("\n===== RIDGE
REGRESSION (TUNED) =====")

param_grid_ridge = {'alpha':
[0.01, 0.1, 1, 10]}

grid_ridge =
GridSearchCV(Ridge(),
param_grid_ridge, cv=5)
grid_ridge.fit(X_train, y_train)

best_ridge =
grid_ridge.best_estimator_

y_pred_ridge =
best_ridge.predict(X_test)
y_pred_ridge_bin = [1 if i > 0.5
else 0 for i in y_pred_ridge]

print("Best Params:",
grid_ridge.best_params_)
print("Accuracy:",
accuracy_score(y_test,
y_pred_ridge_bin) * 100)

#  Confusion Matrix
cm = confusion_matrix(y_test,
y_pred_ridge_bin)
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True,
fmt='d', cmap='Reds',
            xticklabels=['Legitimate',
'Phishing'],
            yticklabels=['Legitimate',
'Phishing'])
plt.title("Confusion Matrix –
Ridge Regression")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

#  Scatter Plot
plt.figure(figsize=(7,6))
plt.scatter(y_test, y_pred_ridge,
alpha=0.4)
plt.axhline(0.5, color='red',
linestyle='--')
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.title("Actual vs Predicted –
Ridge")
plt.show()

#  Feature Importance
coef =
pd.Series(best_ridge.coef_,
index=X.columns)
coef.sort_values(ascending=False
).plot(kind='bar', figsize=(10,6))
plt.title("Feature Importance –
Ridge")
plt.xticks(rotation=90)
```

```
plt.show()

# 
==========================================================================================
# LASSO REGRESSION (TUNED)
# 
==========================================================================================

print("\n===== LASSO REGRESSION (TUNED) =====")

param_grid_lasso = {'alpha': [0.01, 0.1, 1, 10]}

grid_lasso = GridSearchCV(Lasso(), param_grid_lasso, cv=5)
grid_lasso.fit(X_train, y_train)

best_lasso = grid_lasso.best_estimator_

y_pred_lasso = best_lasso.predict(X_test)
y_pred_lasso_bin = [1 if i > 0.5 else 0 for i in y_pred_lasso]

print("Best Params:", grid_lasso.best_params_)
print("Accuracy:", accuracy_score(y_test, y_pred_lasso_bin) * 100)

#  Confusion Matrix
cm = confusion_matrix(y_test, y_pred_lasso_bin)
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Greens',
          xticklabels=['Legitimate', 'Phishing'],
          yticklabels=['Legitimate', 'Phishing'])
plt.title("Confusion Matrix – Lasso Regression")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

#  Scatter Plot
plt.figure(figsize=(7,6))
plt.scatter(y_test, y_pred_lasso, alpha=0.4)
plt.axhline(0.5, color='red', linestyle='--')
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.title("Actual vs Predicted – Lasso")
plt.show()

#  Feature Importance
coef = pd.Series(best_lasso.coef_, index=X.columns)
coef.sort_values(ascending=False).plot(kind='bar', figsize=(10,6))
plt.title("Feature Importance – Lasso")
plt.xticks(rotation=90)
plt.show()
```

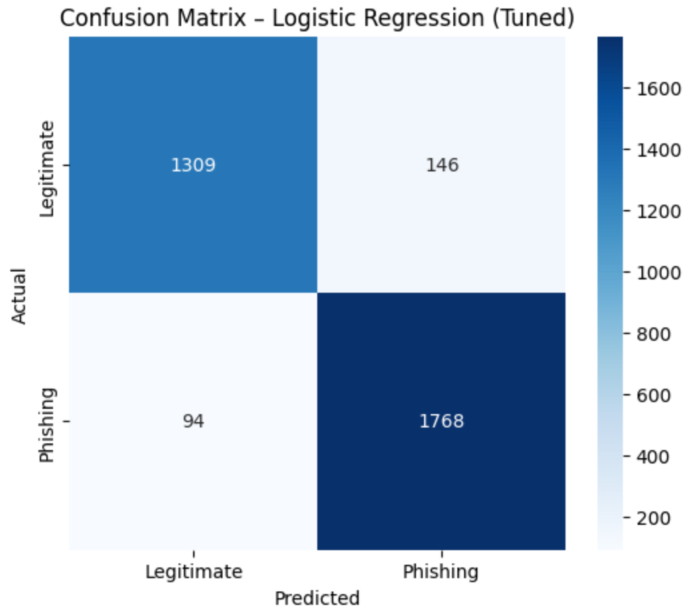
Output

```
🚀 Running ML Models with Hyperparameter Tuning...
📁 Upload your dataset:
```

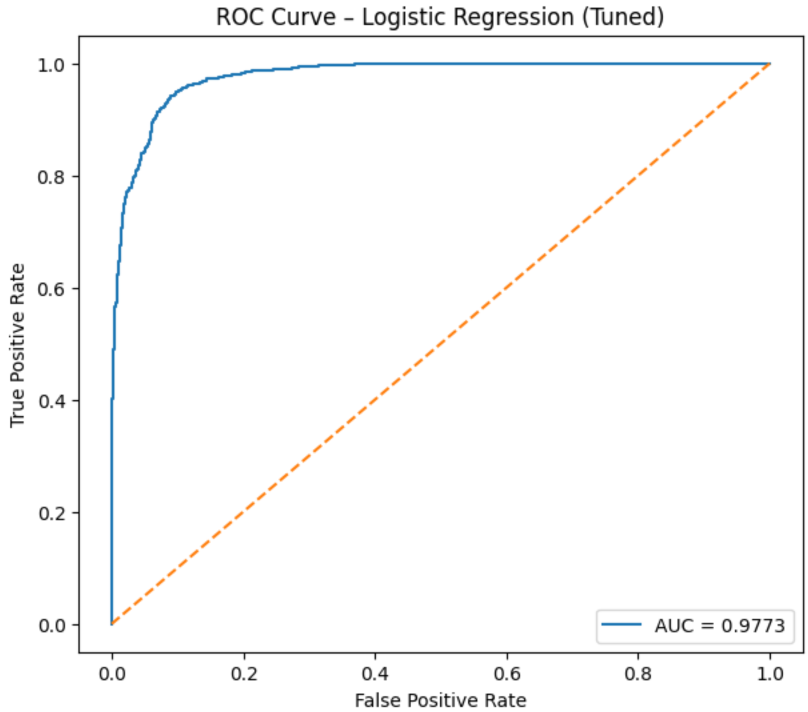
Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving phishing.csv to phishing.csv
✅ Dataset Loaded Successfully
Shape: (11054, 32)
   Index  UsingIP  LongURL  ShortURL  Symbol@  Redirecting//  PrefixSuffix-  \
0      0        1        1         1        1              1             -1
1      1        1        0         1        1              1             -1
2      2        1        0         1        1              1             -1
3      3        1        0        -1        1              1             -1
4      4       -1        0        -1        1             -1             -1

   SubDomains  HTTPS  DomainRegLen  ...  UsingPopupWindow  IframeRedirection  \
0           0      1            -1  ...                 1                  1
1          -1     -1            -1  ...                 1                  1
2          -1     -1             1  ...                 1                  1
3           1      1            -1  ...                -1                  1
4           1      1            -1  ...                 1                  1

   AgeofDomain  DNSRecording  WebsiteTraffic  PageRank  GoogleIndex  \
0           -1            -1               0        -1            1
1            1            -1               1        -1            1
2           -1            -1               1        -1            1
3           -1            -1               0        -1            1
4            1             1               1        -1            1

   LinksPointingToPage  StatsReport  class
0                    1            1     -1
1                    0           -1     -1
2                   -1            1     -1
3                    1            1      1
4                   -1           -1      1

[5 rows x 32 columns]

Missing Values:
 Index                 0
UsingIP               0
LongURL               0
ShortURL              0
Symbol@               0
Redirecting//         0
PrefixSuffix-         0
SubDomains            0
HTTPS                 0
```

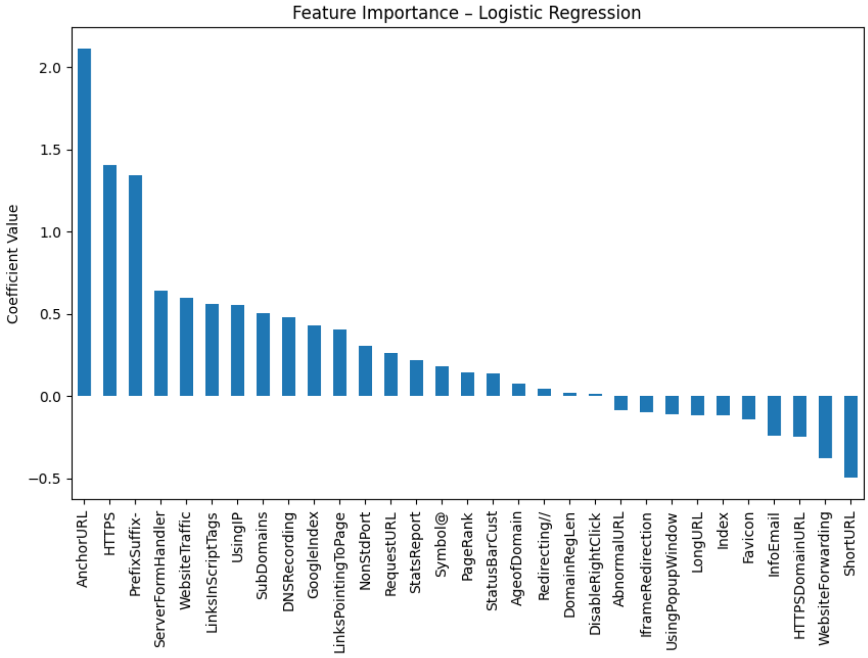
```
DomainRegLen          0
Favicon               0
NonStdPort            0
HTTPSDomainURL        0
RequestURL            0
AnchorURL             0
LinksInScriptTags     0
ServerFormHandler     0
InfoEmail             0
AbnormalURL           0
WebsiteForwarding     0
StatusBarCust         0
DisableRightClick     0
UsingPopupWindow      0
IframeRedirection     0
AgeofDomain           0
DNSRecording          0
WebsiteTraffic        0
PageRank              0
GoogleIndex           0
LinksPointingToPage   0
StatsReport           0
class                 0
dtype: int64

===== LOGISTIC REGRESSION (TUNED) =====
Best Params: {'C': 0.1, 'solver': 'liblinear'}
Accuracy: 92.7645462767561

Classification Report:

              precision    recall  f1-score   support

           0       0.93      0.90      0.92      1455
           1       0.92      0.95      0.94      1862

    accuracy                           0.93      3317
   macro avg       0.93      0.92      0.93      3317
weighted avg       0.93      0.93      0.93      3317
```



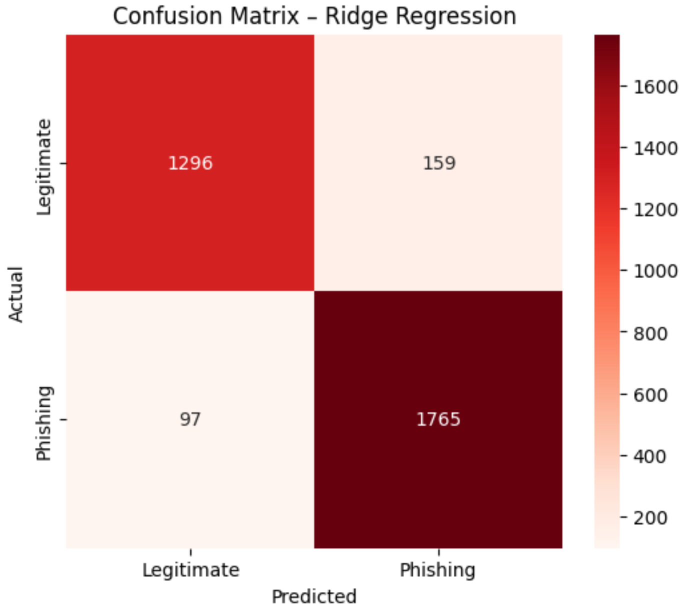
Confusion Matrix – Logistic Regression (Tuned)
Legitimate
Phishing
Actual
1309
146
94
1768
Legitimate
Phishing
Predicted
1600
1400
1200
1000
800
600
400
200

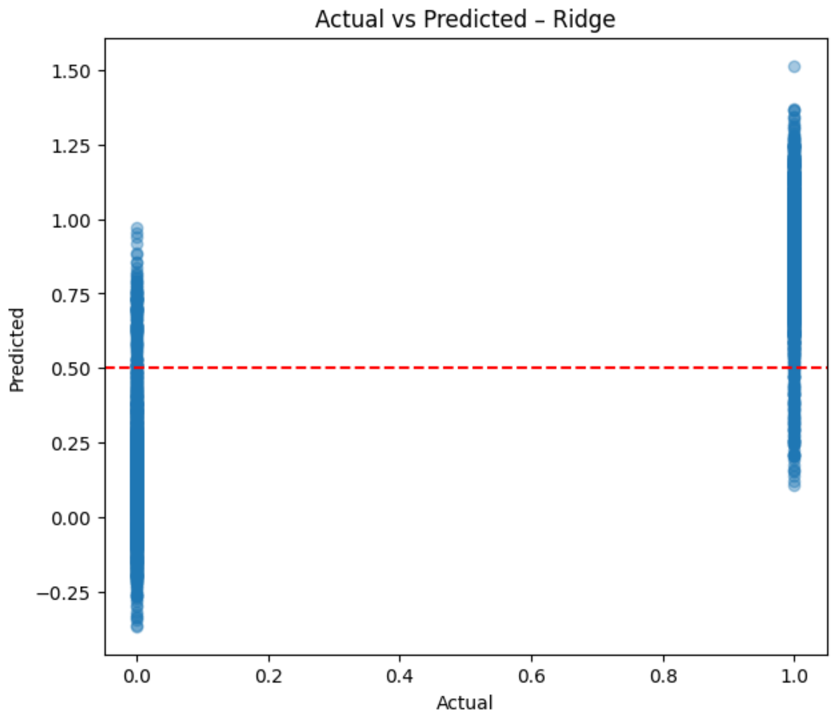


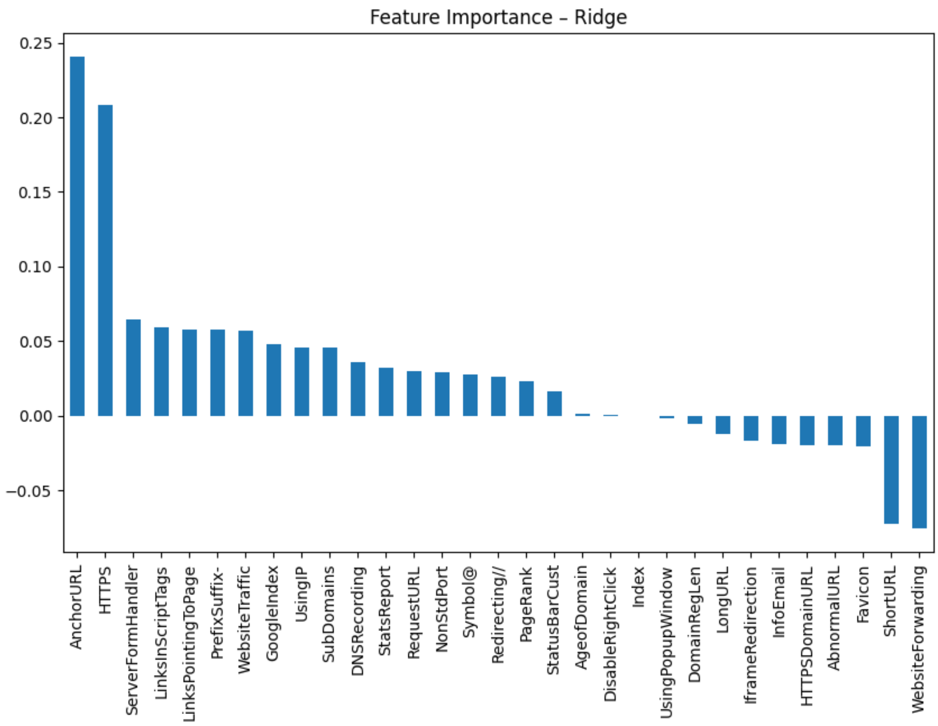
ROC Curve – Logistic Regression (Tuned)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
AUC = 0.9773



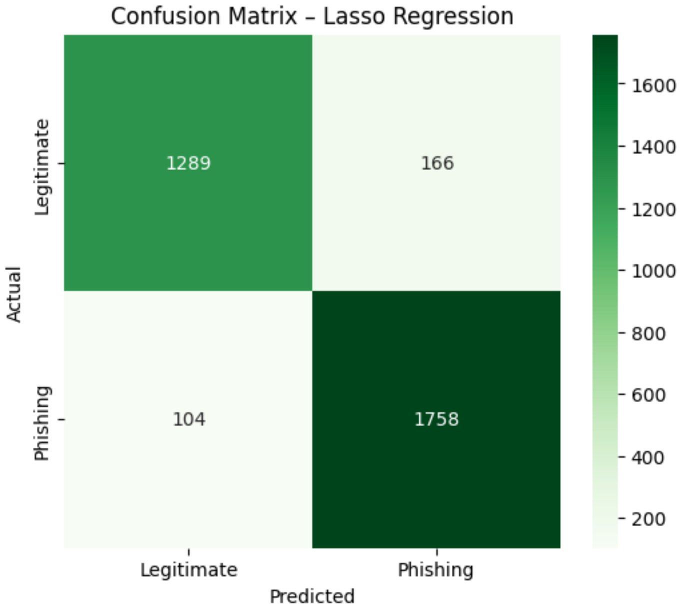

```
===== RIDGE REGRESSION (TUNED) =====
Best Params: {'alpha': 10}
Accuracy: 92.28218269520652
```

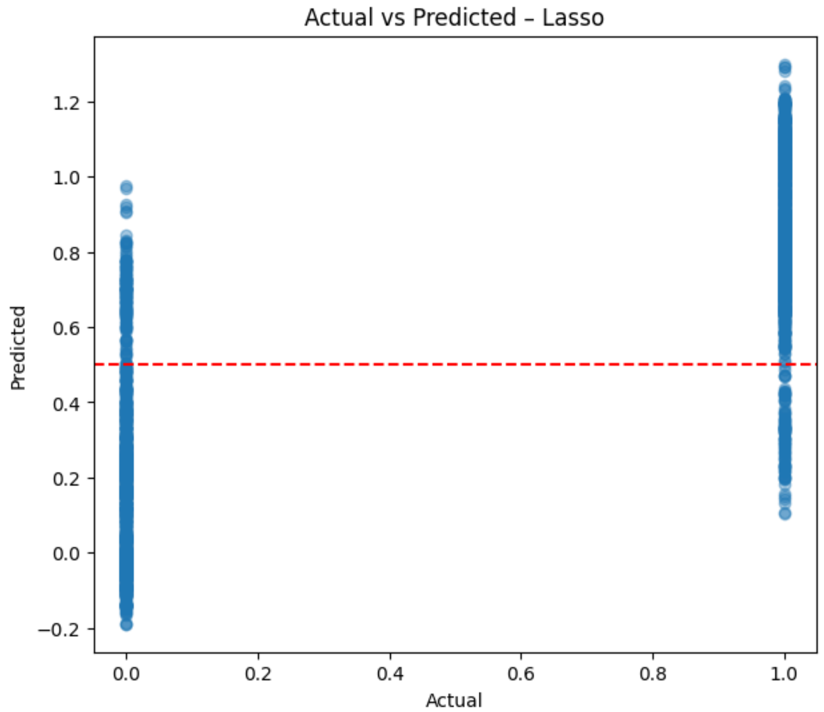


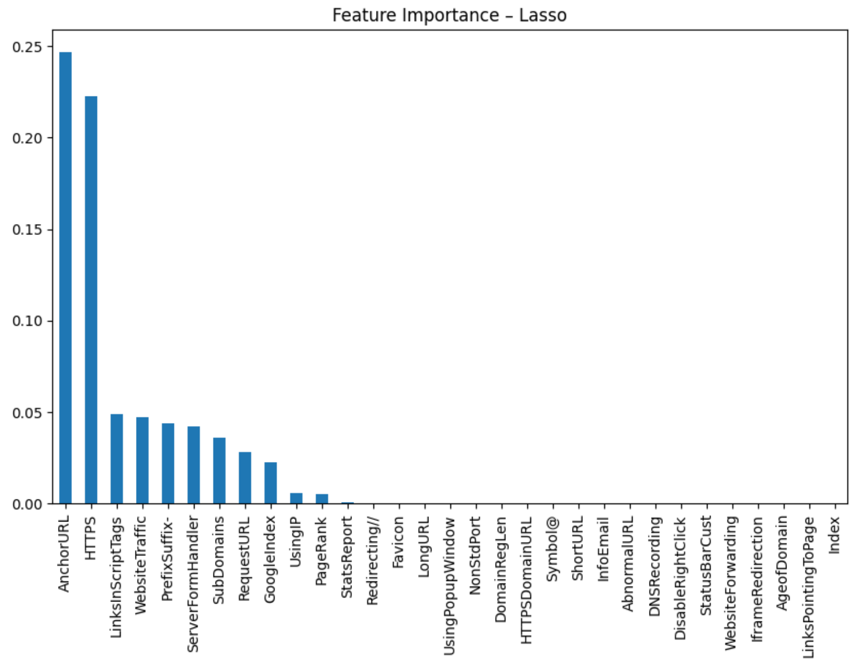

Actual vs Predicted – Ridge
Predicted
Actual
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
0.0
0.2
0.4
0.6
0.8
1.0


Feature Importance – Ridge
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
AnchorURL
HTTPS
ServerFormHandler
LinksInScriptTags
LinksPointingToPage
PrefixSuffix-
WebsiteTraffic
GoogleIndex
UsingIP
SubDomains
DNSRecording
StatsReport
RequestURL
NonStdPort
Symbol@
Redirecting//
PageRank
StatusBarCust
AgeofDomain
DisableRightClick
Index
UsingPopupWindow
DomainRegLen
LongURL
IframeRedirection
InfoEmail
HTTPSDomainURL
AbnormalURL
Favicon
ShortURL
WebsiteForwarding

```
===== LASSO REGRESSION (TUNED) =====
Best Params: {'alpha': 0.01}
Accuracy: 91.86011456135061
```






Feature Importance – Lasso
0.25
0.20
0.15
0.10
0.05
0.00
AnchorURL
HTTPS
LinksInScriptTags
WebsiteTraffic
PrefixSuffix-
ServerFormHandler
SubDomains
RequestURL
GoogleIndex
UsingIP
PageRank
StatsReport
Redirecting//
Favicon
LongURL
UsingPopupWindow
NonStdPort
DomainRegLen
HTTPSDomainURL
Symbol@
ShortURL
InfoEmail
AbnormalURL
DNSRecording
DisableRightClick
StatusBarCust
WebsiteForwarding
IframeRedirection
AgeofDomain
LinksPointingToPage
Index